
Dynamics of Orchestrator–Worker Large Language Model Systems

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Abstract

1 Introduction

Multi-agent LLM frameworks have become a dominant paradigm for complex reasoning tasks, yet little is known about the macroscopic dynamics that emerge when agents communicate through a shared workspace. Prior work has characterised individual agent capabilities [TODO, 2024b], prompt engineering for multi-step reasoning [TODO, 2023], and benchmark performance of collaborative agents [TODO, 2024a], but the dynamical properties of the routing process itself have received less attention.

We approach this gap through the lens of switched dynamical systems [Liberzon, 2003]. Let $x(t) \in \mathbb{R}^d$ denote the workspace embedding at round t . The system evolves as

$$x(t+1) = f_{\sigma(t)}(x(t)), \quad \sigma(t) \in \{R, C, S\}, \quad (1)$$

where $\sigma(t)$ is the routing (switching) signal and f_R, f_C, f_S are the Researcher, Critic, and Synthesizer operators respectively.

Contributions.

1. We formalise orchestrator–worker LLM systems as switched dynamical systems and identify the Synthesizer as a dissipative element.
2. We introduce a suite of routing-sequence metrics (block entropy, motif frequencies, inter-arrival times) and workspace divergence measures applicable to any multi-agent LLM system.
3. We show empirically that LLM orchestrators converge to low-entropy, Synthesizer-free limit cycles, independent of task.

2 Background

Switched dynamical systems. A switched system [Liberzon, 2003] is a family of subsystems $\{f_p\}_{p \in \mathcal{P}}$ together with a switching signal $\sigma : \mathbb{Z}_{\geq 0} \rightarrow \mathcal{P}$ that selects the active subsystem at each step. Stability depends jointly on the subsystem dynamics and on the switching signal; in particular, dwell-time constraints on σ are a standard tool for guaranteeing stability [Liberzon, 2003].

Control-theoretic worker roles. The three workers implement complementary operators on the workspace:

- **Researcher** f_R : expansion — appends new content, $\|x(t+1)\| \geq \|x(t)\|$.
- **Critic** f_C : contraction — appends adversarial commentary opposing prior claims [Lohmiller and Slotine, 1998].
- **Synthesizer** f_S : dissipation/projection — *replaces* the workspace, $x(t+1) = \Pi(x(t))$, $\|\Pi(x)\| \leq \|x\|$ [Khalil, 2002].

The update rule is therefore

$$x(t+1) = \begin{cases} x(t) \oplus \delta_w(t) & w \in \{R, C\} \\ \Pi(x(t)) & w = S, \end{cases} \quad (2)$$

where $\oplus$ denotes text-space concatenation.

3 Methods

Agent architecture. Each worker receives a structured system prompt encoding its role and operates on a shared plain-text workspace via the Anthropic Messages API (claude-haiku-4-5-20251001). The orchestrator implements the feedback policy $\sigma(t) = \pi(x(t), r)$, where r is the user task (reference signal).

Experimental sweep. We sweep over routing strategies $\in \{\text{orchestrator, random, fixed-}w, \text{round_robin}\}$ and Synthesizer placement schedules $\in \{\text{peer, terminal, scheduled-}k\}$. Each cell (task $\times$ placement $\times$ strategy $\times T_o$) is replicated k times with deterministic seeds.

Embeddings. Workspace and final-answer embeddings use all-MiniLM-L6-v2 [Reimers and Gurevych, 2019] ($d = 384$, ℓ_2 -normalised), loaded locally for reproducibility.

4 Results

4.1 Routing entropy

The order- n block entropy of the routing sequence is

$$H_n = - \sum_{w \in \{R, C, S\}^n} p(w) \log_2 p(w), \quad (3)$$

where $p(w)$ is the empirical block frequency pooled across replicates [Cover and Thomas, 2006].

4.2 Motif analysis

We pre-registered eight routing motifs *prior* to data collection (Table 1) to avoid post-hoc inflation of significance.

Table 1: Pre-registered routing motifs and observed mean counts per replicate.

Motif	Interpretation	Orchestrator / Random
RC	Researcher–Critic alternation, period 2	
RCRC	Period-2 limit cycle	
RCSRCS	Three-phase rhythm (conjectured healthy)	
RSRS	Premature consolidation	
SS	Synthesizer doublet	
SSSS	Degenerate self-feeding	
RR	Researcher monoculture	
CC	Critic monoculture	

4.3 Workspace divergence

For K replicates, the finite-time divergence-rate proxy at round t is

$$\Delta(t) = \frac{1}{\binom{K}{2}} \sum_{i < j} (1 - \cos(x_i(t), x_j(t))). \quad (4)$$

Increasing $\Delta(t)$ indicates sensitive dependence on the random seed, analogous to the finite-time Lyapunov exponent [Strogatz, 2015].

5 Discussion

6 Conclusion

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- TODO. Todo: multi-agent llm benchmark paper. *TODO*, 2024a.
- TODO. Todo: single-agent llm capability paper. *TODO*, 2024b.